RAJENDRA PRASAD N

LEAD MULTI-CLOUD & AI INFRASTRUCTURE ENGINEER | AWS, AZURE, TERRAFORM, KUBERNETES (EKS/AKS) & GENAI

rajendran.scm@gmail.com
+1 (469) 926-4201
Dallas, TX
linkedin.com/in/rajendranelakurthi
rajendranelakurthi.github.io

Lead Multi-Cloud & AI Infrastructure Engineer with 10 years of hands-on experience designing, provisioning, securing, and operating enterprise cloud platforms across AWS and Microsoft Azure. Deep active coding proficiency in Infrastructure as Code using **Terraform, Azure Bicep/ARM, and CloudFormation/CDK**, paired with cloud automation tooling built in **Python (Boto3, Azure SDK)** and Bash. Proven track record architecting and scaling production **Kubernetes environments (EKS and AKS)**, enterprise multi-tier networking (VPCs, VNets, Transit Gateways, VPN Gateways), and zero-trust IAM least-privilege governance. Proven experience architecting enterprise infrastructure for **GenAI and Agentic AI workloads**—including LLM hosting, private inference routing, vector databases (Milvus, pgvector), RAG pipelines, **AWS Bedrock, Azure OpenAI Service**, and multi-agent coordination frameworks (LangGraph, AutoGen). Strong advocate for FinOps cost efficiency, data governance, model risk compliance, end-to-end observability, with hands-on background in **enterprise data pipelines (Databricks)** and **Java development**.

CERTIFICATIONS

AWS Certified Solutions Architect - Associate

Microsoft Certified: Azure DevOps Engineer Expert

Microsoft Certified: Azure Fundamentals

GitLab Certified Associate

MuleSoft Certified Developer - Level 1

MULTI-CLOUD & AI INFRASTRUCTURE HIGHLIGHTS

- **Multi-Cloud IaC & Automation** Architected, provisioned, and operated enterprise infrastructure across AWS and Azure using modular Terraform and Azure Bicep, standardizing automated state management, CI/CD validation, and drift detection.
- **GenAI & Agentic AI Infrastructure** Engineered high-throughput, secure infrastructure for enterprise GenAI platforms—deploying vector databases (Milvus), RAG pipelines, AWS Bedrock, and Azure OpenAI with private endpoints, model governance, and multi-agent frameworks.
- **Kubernetes & Enterprise Networking** Scaled and hardened production EKS and AKS clusters with Karpenter, Helm, and GitOps; implemented multi-tier networking spanning VPCs, Azure VNets, Transit Gateways, ExpressRoute, and VPN Gateways.
- **Governance, FinOps & Observability** Enforced zero-trust IAM least-privilege, automated secrets rotation, model risk compliance logging, FinOps compute/GPU cost optimization, and end-to-end telemetry with CloudWatch, Azure Monitor, and Prometheus.

CORE TECHNOLOGIES

CLOUD PLATFORMS & MULTI-CLOUD

AWS (EC2, S3, EKS, Lambda, Bedrock, RDS, VPC, Transit Gateway, IAM, KMS), Microsoft Azure (VMs, AKS, Azure OpenAI, Azure AI Search, Blob Storage, VNets, VPN Gateway, Entra ID, Key Vault)

INFRASTRUCTURE AS CODE (IAC)

Terraform (modular design, workspaces, state locking, AWS/AzureRM providers), Azure Bicep, ARM Templates, CloudFormation, AWS CDK, Checkov, tfsec

GENAI & AGENTIC AI INFRASTRUCTURE

AWS Bedrock, Azure OpenAI Service, LLM Hosting & Inference, RAG Pipelines, Vector Databases (Milvus, pgvector, Azure AI Search), Multi-Agent Coordination Frameworks (LangGraph, AutoGen, CrewAI), Semantic Caching, GPU Node Pools

CONTAINERS & ORCHESTRATION

Docker, Kubernetes, Amazon EKS, Azure Kubernetes Service (AKS), Helm, Argo CD, GitOps, Karpenter, Cluster Autoscaler, Calico, K8s RBAC, Workload Identity

ENTERPRISE NETWORKING & SECURITY

AWS VPC, Azure VNet (Hub-and-Spoke), Transit Gateways (TGW), VPN Gateways, Direct Connect, ExpressRoute, PrivateLink, Private Endpoints, IAM Least Privilege, Azure Entra ID / RBAC, KMS, Azure Key Vault, Secrets Manager

AUTOMATION, SCRIPTING & APIS

Python (Boto3, Azure SDK, FastAPI), Bash, PowerShell, HCL, REST APIs, JSON/YAML, cloud operational runbooks, automated provisioning CLI tooling

OBSERVABILITY, GOVERNANCE & FINOPS

CloudWatch, Azure Monitor, Log Analytics, Prometheus, Grafana, OpenTelemetry, Cost Allocation Tagging, AWS Cost Explorer, Azure Cost Management, FinOps, Model Risk Compliance, Data Governance

CI/CD, DATA PIPELINES & JAVA

GitHub Actions, GitLab CI, AWS CodePipeline, Jenkins, Databricks on AWS/Azure, Apache Spark, Java (Maven, Spring Boot, Tomcat/JBoss, JVM memory tuning), SonarQube, JFrog Artifactory

PROFESSIONAL EXPERIENCE

Lead Cloud Infrastructure & DevOps Engineer | AT&T Services Inc.

08/2023 - Present

Skills Used: AWS, Azure, Terraform, Azure Bicep, Python (Boto3, Azure SDK), EKS, AKS, AWS Bedrock, Azure OpenAI, Milvus, Databricks, Helm, GitHub Actions

- Actively designed, provisioned, and operated enterprise multi-cloud infrastructure across AWS and Microsoft Azure using modular Terraform and Azure Bicep, standardizing automated state locking, CI/CD validation, and drift remediation.
- Architected underlying infrastructure for enterprise GenAI and Agentic AI workloads: deployed and scaled high-throughput vector databases (Milvus, pgvector) and RAG retrieval pipelines on EKS and AKS, provisioning GPU-accelerated node pools and low-latency inference routing.
- Integrated and secured GenAI cloud services including AWS Bedrock and Azure OpenAI Service, establishing private connectivity via AWS PrivateLink and Azure Private Endpoints, API rate-limiting orchestration layers, and prompt/response audit telemetry for model risk compliance.
- Built foundational cloud infrastructure to support multi-agent coordination frameworks (LangGraph, AutoGen), provisioning event-driven messaging queues (SQS, Azure Event Hubs), state persistence stores, and distributed worker nodes for agentic execution loops.
- Developed operational tooling and automation scripts against AWS and Azure APIs in Python (utilizing Boto3 and Azure SDK) to automate multi-account resource provisioning, identity governance checks, and certificate lifecycle management.
- Automated provisioning and network isolation for Databricks environments across AWS and Azure accounts, standardizing VPC/VNet peering, IAM least-privilege roles, storage access policies, and enterprise data pipeline integration.
- Designed and implemented secure multi-tier networking connecting on-premises datacenters, AWS VPCs, and Azure VNets utilizing AWS Transit Gateways, Azure VPN Gateways, Route 53, and network security groups (NSGs).
- Enforced end-to-end data governance, model risk compliance, and FinOps cost efficiency—leveraging Karpenter autoscaling, right-sizing GPU/compute instances, and centralizing observability across Amazon CloudWatch, Azure Monitor, and Prometheus.

Senior Cloud & DevOps Engineer | GlobalLogic

03/2023 - 08/2023

Skills Used: Azure, AWS, AKS, EKS, Terraform, Azure Bicep, Python, GitLab CI, Docker, Kubernetes, Trivy

- Set up, scaled, and secured production Kubernetes environments across Azure Kubernetes Service (AKS) and Amazon EKS using Terraform and Helm, standardizing ingress controllers, network policies, and horizontal pod autoscaling (HPA).
- Authored Python automation scripts and GitLab CI pipelines to streamline container security scanning (Trivy), image signing, and artifact promotion across multi-cloud container registries.
- Implemented IAM least-privilege access governance, establishing role-based access control (RBAC), Azure Entra ID / AD service principals, and OIDC federated authentication for automated deployment runners.
- Optimized runtime environments for containerized Java microservices on Kubernetes, configuring JVM memory parameters, readiness/liveness probes, and zero-downtime rolling updates.
- Managed Linux and Windows auto-scaling build agents and migrated legacy Jenkins workflows to GitLab CI, improving pipeline throughput and deployment reliability.

Senior Cloud & DevOps Engineer II | Akamai Technologies

12/2021 - 03/2023

Skills Used: AWS, Terraform, CloudFormation, Python (Boto3), EKS, Transit Gateway, VPN, CloudWatch, GitHub Actions

- Provisioned and operated scalable AWS cloud infrastructure using Terraform, CloudFormation, and GitHub Actions, establishing reusable IaC modules and automated multi-environment promotion.
- Implemented secure multi-tier networking including VPCs, subnets, route tables, AWS Transit Gateways, VPN gateways, and Direct Connect for resilient hybrid enterprise connectivity.
- Built Python (Boto3) and Bash automation scripts to interact with AWS APIs, automating resource tagging, orphaned volume cleanup, and security compliance audits.
- Architected cross-region disaster recovery and backup automation using AWS Elastic Disaster Recovery and AWS Backup, maintaining high availability for mission-critical services.
- Deployed containerized microservices to Amazon EKS using Helm charts, configuring cluster security controls, secrets integration, and centralized CloudWatch logging and metrics.

Senior DevOps Engineer / Senior SRE | Turvo India Pvt Ltd

04/2020 - 12/2021

Skills Used: AWS, Terraform, Python (FastAPI), Kubernetes, Prometheus, Grafana, Jenkins, Secrets Manager, Java

- Designed and delivered a Terraform and Kubernetes infrastructure platform for high-volume logistics and real-time payment transactions, enforcing automated provisioning and container orchestration.
- Developed internal platform automation APIs in Python (FastAPI) to integrate Jenkins, Jira, and cloud provider APIs, standardizing deployment verification and operational self-service.
- Enforced IAM least-privilege governance, MFA enforcement, and secure secrets management utilizing AWS Secrets Manager and KMS encryption across multi-account environments.
- Developed custom Prometheus exporters in Python to capture application and messaging queue latency metrics, building Grafana operational dashboards for SRE monitoring and incident response.
- Supported enterprise backend services and data pipelines developed in Java, assisting development teams with JVM profiling, connection pooling, and low-latency database connectivity.

Senior Cloud & DevOps Engineer | HCL Technologies

02/2019 - 04/2020

Client: Mercedes-Benz Research & Development India

Skills Used: AWS, CloudFormation, Java, Maven, Jenkins, Ansible, RDS, EC2, CloudWatch, KMS

- Provisioned and supported enterprise cloud infrastructure for connected-vehicle platforms on AWS using CloudFormation, EC2, RDS, ALB, and CloudFront.
- Partnered with enterprise Java development teams to build and optimize Maven CI/CD pipelines in Jenkins, configuring Tomcat and JBoss application runtime environments.
- Automated fleet-level instance configuration and patch management for Linux and Windows workloads using Ansible playbooks and PowerShell scripting.
- Executed cloud migration initiatives, moving on-premises workloads to AWS VPCs with automated provisioning templates, least-privilege IAM policies, and KMS encryption.

DevOps & Infrastructure Engineer | Mphasis Ltd

08/2016 - 12/2018

Client: Hewlett Packard Enterprise (HPE)

Skills Used: AWS, Terraform, Python, Shell, Ansible, Jenkins, Maven, Splunk, Nexus

- Authored and maintained Terraform modules and Ansible playbooks to provision and configure multi-tier application infrastructure across development and staging environments.
- Built automated deployment pipelines using Jenkins, Git, Maven, and Nexus to package and deploy enterprise Java application components.
- Developed Python and Shell operational scripts to automate administrative tasks, cloud configuration updates, and log rotation.
- Configured centralized logging and operational monitoring with Splunk and CloudWatch to analyze deployment metrics and accelerate incident resolution.

EDUCATION

B.Tech (2009-2013) in Information Technology

Prakasam Engineering College, Jawaharlal Nehru Technological University, Kakinada